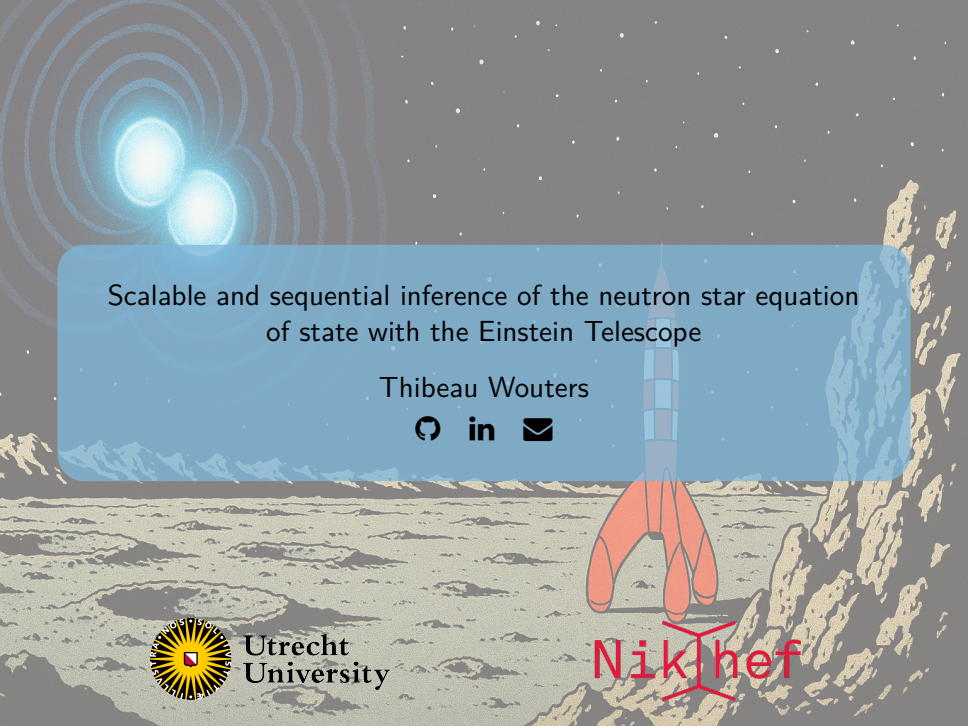




Scalable and sequential inference of the neutron star equation of state with the Einstein Telescope

Thibaud Wouters



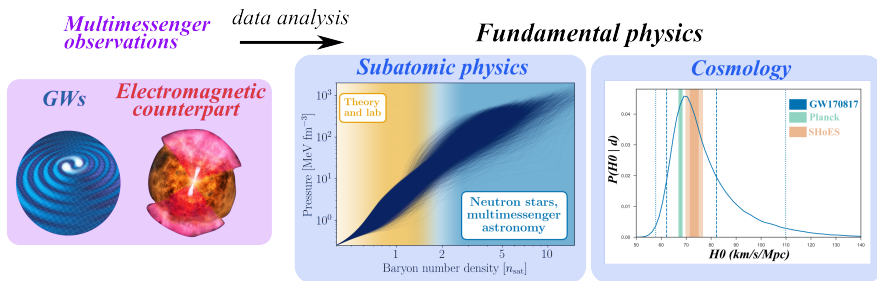
Utrecht
University



Neutron stars and fundamental physics

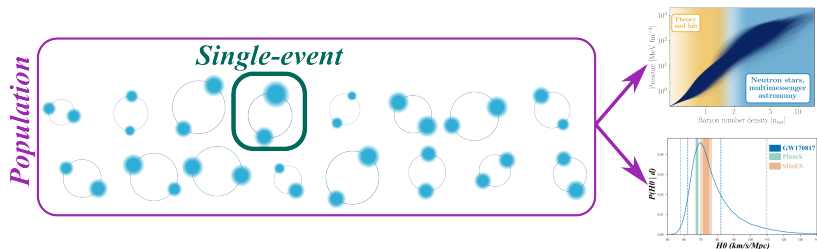
Neutron star mergers and multimessenger observations probe

- Equation of state of ultradense matter [1]
- Expansion of the universe (H_0) [2]
- Tests of GR, populations, nucleosynthesis,...

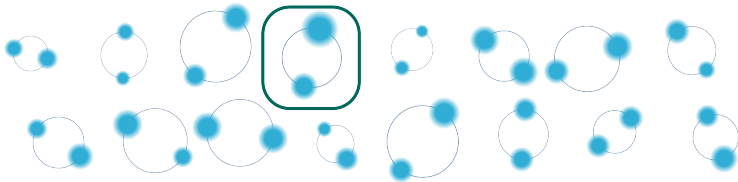


Challenges for Einstein Telescope

- Einstein Telescope
 - $\sim 30,000$ binary neutron star mergers/year
 - ~ 100 with detected electromagnetic counterpart
- **Challenge:** Data analysis (Bayesian inference) needs to scale up:
 - **Single-event:** analyze each event independently
 - **Population:** combine events into science insights
- **Solution:** GPU-acceleration $\times$ algorithm innovations



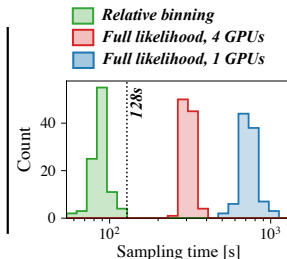
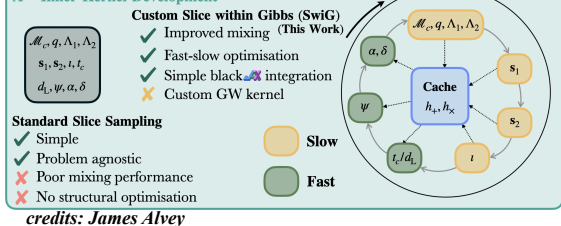
Single-event



Single event: slice-within-Gibbs sampler (2607.28265)

- Waveform generation is expensive
- **Slice-within-Gibbs** [3]: move in ‘blocks’
 - **Slow**: new waveform generations, cached
 - **Fast**: projections, reuse cache
- Inference (for LVK signals!) in 15 mins with **no** approximation, or faster than signal duration (128 s) with relative binning

A – Inner Kernel Development



Towards ET with the BeNL community

- Development extended towards ET signals
 - RIPPLE 🗓: state-of-the-art waveforms (**talk by Robin**)
 - JIM 🗓: developing samplers (Thomas C. K. Ng)
- Strong BeNL presence (Harsh Narola, Peter T. H. Pang, Fabian Gittins, Isaac C. F. Wong, . . .)

New hardware-accelerated and accurate gravitational waveform approximants for fast parameter estimation

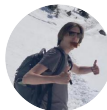
🗓 Oct 9, 2026, 11:50 AM

🕒 20m

📍 Aula Magna

+ *Jef Heynen, Mick Wright,
Justin Janquart for lensing!*

Robin Chan



RIPPLE

Thomas C. K. Ng



JIM

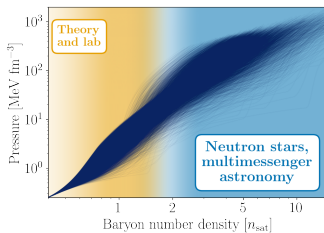
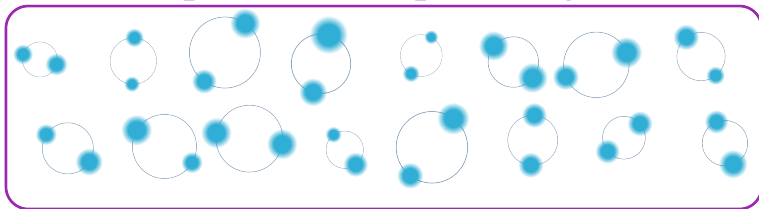
GPU-accelerated Bayesian inference for diffracted gravitational wave signals

🗓 Oct 8, 2026, 5:50 PM

🕒 20m

📍 Aula Magna

Population $\rightarrow$ equation of state

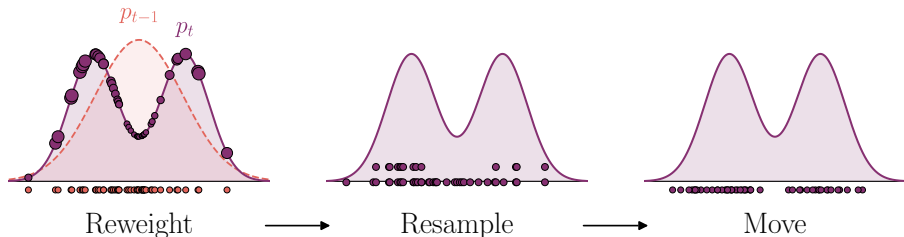


Equation of state inference with JESTER

- Target: Equation of state (EOS) posterior from N events d_i

$$\mathcal{P}(\theta_{\text{EOS}}|\{d_1, \dots, d_N\}) \propto \mathcal{L}(\{d_1, \dots, d_N\}|\theta_{\text{EOS}}) \times \pi(\theta_{\text{EOS}})$$

- Sequential Monte Carlo**: particles from prior to posterior through a sequence of intermediate densities p_t
- JESTER  [4]: used by LVK and ET



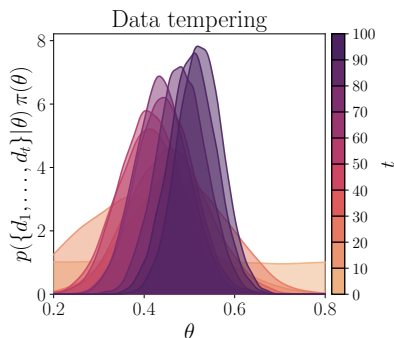
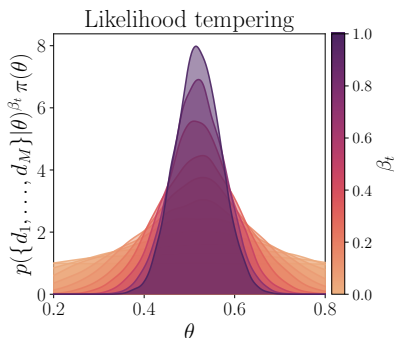
Likelihood tempering vs. data tempering

1 Likelihood tempering:

$$p_t(\theta) \propto \mathcal{L}(\{d_1, \dots, d_N\}|\theta)^{\beta_t} \pi(\theta) \quad \beta_t : 0 \rightarrow 1$$

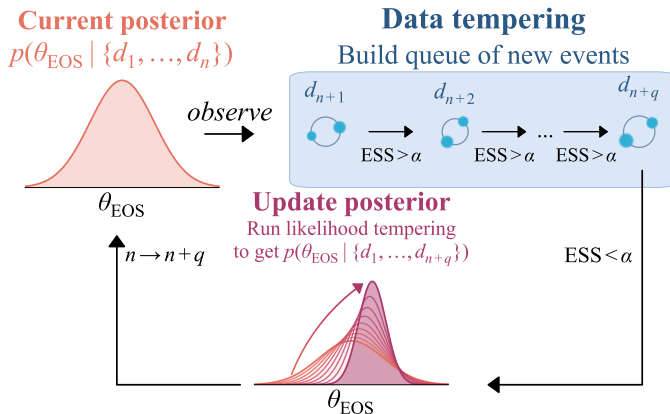
2 Data tempering:

$$p_t(\theta) \propto \mathcal{L}(\{d_1, \dots, d_t\}|\theta) \pi(\theta) \quad t : 1 \rightarrow N$$



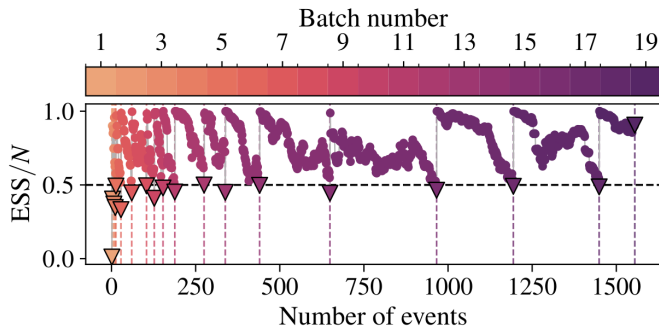
Adaptive hybrid algorithm (2610.07975)

- ET catalogs will grow rapidly: sequentially update posteriors [5]
- Effective sample size (ESS): metric for adaptive batching
- Use likelihood tempering for robustness



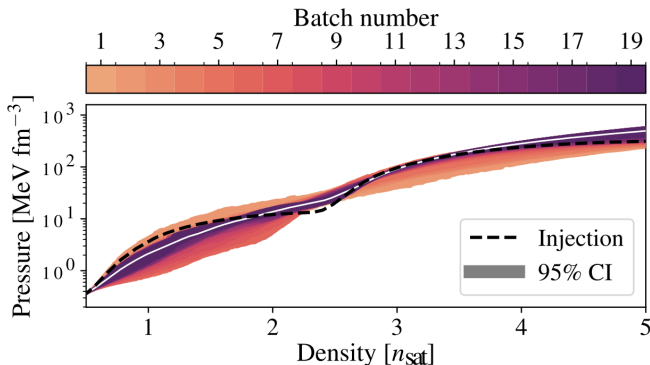
Proof of concept (2610.07975)

- Toy catalogue: 1 month of binary neutron stars (GW) (~ 1500)
- Runtime ~ 2 hrs; expectation for 1 year: ~ 2 days
- Processing events 'in real time', as we will do it ~ 10 years from now!

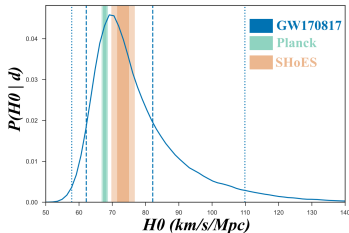
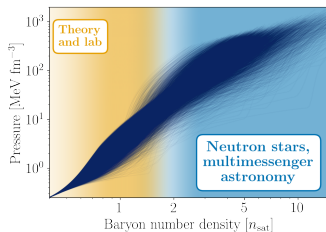
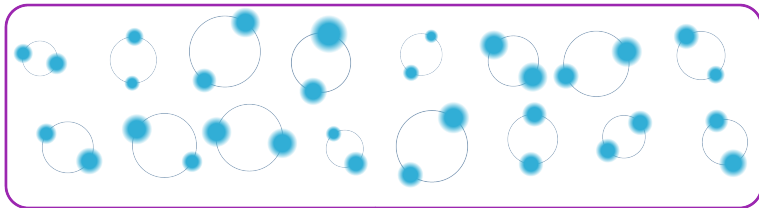


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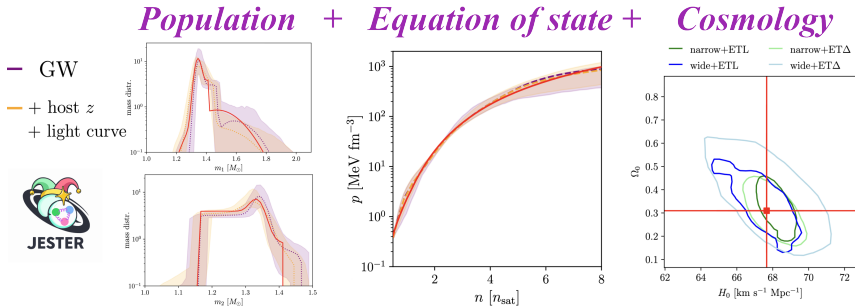


Population $\rightarrow$ *equation of state* + *cosmology*



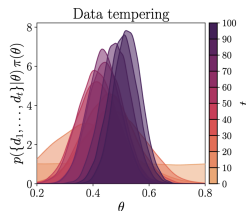
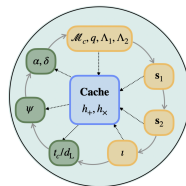
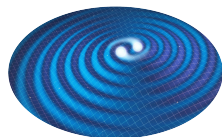
Multimessenger forecasts (2607.28438)

- Multimessenger forecasts (1 year of mergers) for population + equation of state + cosmology [6] with JESTER
- GW + kilonova + GRB afterglow on CPU (NMMA 🌀)
- Single-event analysis becoming the bottleneck



Conclusions

- **Challenge:** ET will see $\mathcal{O}(10^4)$ binary neutron star mergers per year
- **Solution:** GPU $\times$ algorithmic innovations
 - **Single-event:** slice-within-Gibbs
 - **Population:** sequential Monte Carlo
- **Results:**
 - **ripple, jim:**
 - LVK signals (and lensing!)
 - ET under development
 - **jester**
 - Scale to $\mathcal{O}(10^3)$ sources
 - Multimessenger forecasts
- Strong BeNL presence!



Thanks for listening!

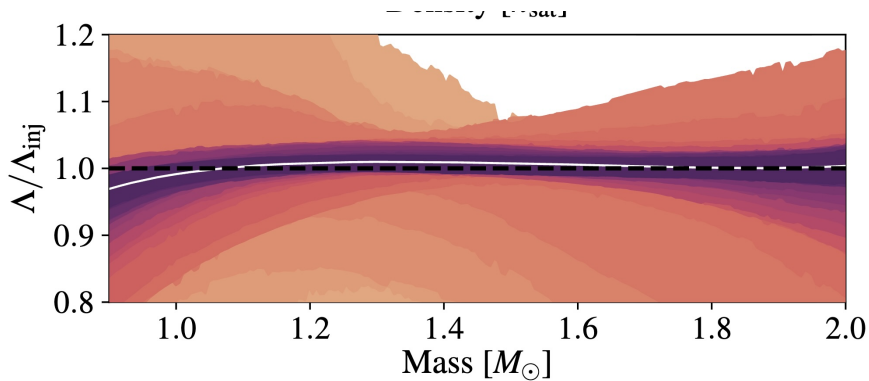


References

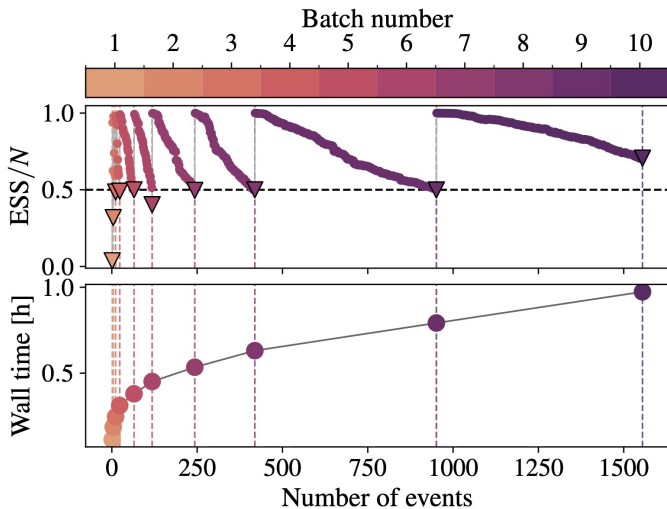
- [1] B. P. Abbott et al. “GW170817: Measurements of neutron star radii and equation of state”. In: *Phys. Rev. Lett.* 121.16 (2018), p. 161101. DOI: [10.1103/PhysRevLett.121.161101](https://doi.org/10.1103/PhysRevLett.121.161101). arXiv: [1805.11581](https://arxiv.org/abs/1805.11581) [gr-qc].
- [2] B. P. Abbott et al. “A gravitational-wave standard siren measurement of the Hubble constant”. In: *Nature* 551.7678 (2017), pp. 85–88. DOI: [10.1038/nature24471](https://doi.org/10.1038/nature24471). arXiv: [1710.05835](https://arxiv.org/abs/1710.05835) [astro-ph.CO].
- [3] David Yallup et al. “Ab Initio Real-Time Gravitational-Wave Parameter Estimation”. In: (July 2026). arXiv: [2607.28265](https://arxiv.org/abs/2607.28265) [gr-qc].
- [4] Thibaud Wouters et al. “Leveraging differentiable programming in the inverse problem of neutron stars”. In: *Phys. Rev. D* 112.4 (2025), p. 043037. DOI: [10.1103/v2y8-kxvx](https://doi.org/10.1103/v2y8-kxvx). arXiv: [2504.15893](https://arxiv.org/abs/2504.15893) [astro-ph.HE].
- [5] Thibaud Wouters et al. *Scalable and sequential inference of the neutron star equation of state with the Einstein Telescope*. 2026. arXiv: [2610.07975](https://arxiv.org/abs/2610.07975) [astro-ph.HE]. URL: <https://arxiv.org/abs/2610.07975>.
- [6] Hauke Koehn et al. “Binary neutron stars in the next-generation era: Multi-messenger detection prospects and constraints on the equation of state, mass distribution, and cosmology”. In: (July 2026). arXiv: [2607.28438](https://arxiv.org/abs/2607.28438) [astro-ph.HE].

APPENDIX

Λ recovery



Zero noise results



Zero noise results

